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The making and breaking of minds: How social interactions shape the human mind, by Isabella Sarto-Jackson, Vernon Press, DE, USA, 2022, 278 pp., USD \$72 (hardcover), ISBN: 978-1-62273-331-6

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To cite this article: Elena Walsh (2023): How to 'make or break' a mind: causes and causal difference-makers in developmental psychology, Philosophical Psychology, DOI: [10.1080/09515089.2023.2193242](https://doi.org/10.1080/09515089.2023.2193242)

To link to this article: <https://doi.org/10.1080/09515089.2023.2193242>



Published online: 30 Mar 2023.



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BOOK REVIEW

How to 'make or break' a mind: causes and causal difference-makers in developmental psychology

The making and breaking of minds: How social interactions shape the human mind, by Isabella Sarto-Jackson, Vernon Press, DE, USA, 2022, 278 pp., USD \$72 (hardcover), ISBN: 978-1-62273-331-6

Sarto-Jackson's *The Making and Breaking of Minds* is a rich and original contribution to the fields of developmental psychology and preventative medicine. Its central claim is that the social environments we encounter in childhood and adolescence have a profound and historically under-appreciated impact on brain development and cognitive function in adulthood, predicting a wide range of health and behavioral outcomes. A particular focus is placed on the role of poverty, trauma, and neglect in childhood on later life outcomes (in particular mental illness, and propensity toward violence and antisociality).

Since the landmark adverse childhood experiences (ACEs) study by Felitti et al. (1998), which linked traumatic childhood events to health outcomes in later life, a research movement has grown supporting the building of trauma-informed institutions and programs promoting resilience. Despite this, as Sarto-Jackson herself reports, there is a remarkable lack of communication between researchers in the biological sciences and sciences of mind, on the one hand, and researchers who study the social determinants of health and wellbeing, on the other. Sarto-Jackson, herself a neuroscientist, hopes that her monograph will offer a pathway to bridge this gap.

In my view, she partially succeeds in this aim. The monograph, as I see it, makes two important and novel contributions to the literature that assist developing an interdisciplinary approach toward studying the social determinants of health and wellbeing. The first is Sarto-Jackson's construal of cognitive development as an adaptation to a social niche, and the second is her account of some of the mechanisms likely involved in linking features of early social environments to later life outcomes. Both of these contributions are summarized below. I think the central limitation of the monograph is that it lacks an account of how the various descriptive levels (genetic, neurobiological, cognitive, psychological, social, evolutionary) discussed throughout inter-relate as part of a general causal framework for explaining the outcomes of interest to the author. After providing a thematic overview of the monograph, I explain why providing a causal framework of this sort is desirable.

The first notable contribution of the monograph is Sarto-Jackson's construal of cognitive development as an adaptation to a social niche, and her illustration of how this view pushes against the historical dominance of preformationist and deterministic assumptions about human nature. As Sarto-Jackson recounts in Chapter 1, preformationism, in its earlier seventeenth and eighteenth century incarnations, held that preformed miniature humans (homunculi) existed in male semen or female eggs and needed to be "unwrapped" to eventually grow into full-size humans. In the early

twentieth century, this idea was superseded by the new discipline of biochemistry, within which the structure of DNA was elucidated and the metaphor of a “genetic program” first emerged (P. Griffiths & Stotz, 2013). This metaphor is preformationist in assuming a simple, unmediated causal relationship between developmental information encoded in genes and a given phenotype (Nicholson, 2014).

In Chapter 2, Sarto-Jackson describes the nineteenth- and twentieth-century view that the brain was “hardwired” and that various roughly demarcated brain functions (e.g., language, emotion) were localized to particular brain regions. She then charts the development of the concept of neuroplasticity, which describes the capacity of the human brain to reorganize itself and flexibly adjust to environmental conditions across the lifespan. Sarto-Jackson shows how neuroplasticity provides the basis for wide-ranging learning and memory processes that are particularly profuse during childhood and adolescence. She also notes that the exceptional malleability of the developing brain leaves it highly vulnerable to negative impact from its surroundings.

From Chapter 3 onwards, Sarto-Jackson presents a history of key findings across the biological and social sciences that precipitated movement toward a non-deterministic approach to human development that focuses on the important role played by social environments. Here she makes the key point that our “social niche” modulates brain development, neuroplastic processes, and genetic expression, and that harmful or absurd conclusions can follow from deterministic explanatory frameworks that ignore social context. Just one example is Sarto-Jackson’s discussion of the “dromomania” diagnosis, used well into the twentieth century, denoting an “irrational impulse to wander or travel without purpose”. This diagnosis was applied both to homeless people and to children fleeing their own homes. Sarto-Jackson discusses the case of a young boy called Ludvig E., who was diagnosed with “inborn vagabondage” (pp. 55–57). This deterministic explanation obscured the fact that the boy seemed to in fact be fleeing home to avoid being physically beaten (see pp. 55–57). In this case, what looked like obviously fitness-enhancing behavior was pathologised.

Sarto-Jackson characterizes the late twentieth-century as a period in which the social environment was recognized as a relevant and important explanatory factor for “deviant” behavior. She argues that despite this, deterministic explanations of human behavior continued to manifest. In Chapter 4, Sarto-Jackson discusses the 2009 Waldroup trial. Waldroup, who had shot and mutilated his ex-wife, was on trial for attempted murder. It was found that Waldroup’s DNA contained a certain allelic variant of the monoamine oxidase-A (MAO-A) gene, thought to predict aggression. The media sensationalized the idea of a “murder” or “warrior” gene that predicted violence, and one juror described Waldroup as simply suffering from the “bad gene” (p. 86). In the end, Waldroup’s sentence was reduced from attempted murder to manslaughter.

In the second half of the book, Sarto-Jackson makes a second important contribution to understanding how early social environments influence long-term health and behavioral outcomes. She does this by providing a detailed synthesis of recent work in neuroscience and cognitive science demonstrating that memory and emotion are closely coupled systems, contrasting this new approach against nineteenth- and twentieth-century “faculty psychology” approaches, which saw the two as distinct and separable. In these chapters, Sarto-Jackson uses the coupling of memory and emotion as a basis for claiming that traumatized people live in “an emotionally relabeled world in which the evaluation of social cues is biased toward the negative”, leading to a perpetual state of high arousal (p. 159). Sarto-Jackson describes this as the functional consequence of trauma in adulthood, as well as the “more dramatic neuroplastic restructuring”

(p. 162) that occurs in childhood trauma. By describing some of the cognitive and neurobiological mechanisms that are transformed by social contexts and in turn influence behavior, Sarto-Jackson paves the way for sustained dialogue between cognitive scientists and neuroscientists on the one hand, and theorists who focus on social environments (e.g., social psychologists, social scientists, clinical psychologists).

Chapters 7 and 8 directly discusses the harmful effects of neglectful, abusive, and violent parenting. Sarto-Jackson shows how twentieth-century experiments involving rhesus monkeys, rats and rat pups, primates, and observations of human child-rearing gradually led to an understanding of the far-reaching impacts of early maltreatment on adult temperament. She also describes similar consequences associated with socio-economic deprivation (especially poverty). A central point developed in these later chapters is the idea that niche construction (Odling-Smee et al., 1996) can be deployed to explain the complex phenomenon of intergenerational violence. Her basic claim is that people will recreate a familiar social niche rather than adapt to a new one:

Alarminglly, if external environments that are available do not match internal, mental, and emotional structures, individuals will act to manipulate their environment. (p. 224)

Changing the external environment to “match” an “internal structure” is seen through the lens of Bowlby’s (1951) mid-twentieth century work on attachment theory. An infant with a neglecting or abusive mother will develop “disorganized attachment”, characterized by attempts to both anxiously attach to as well as avoid the parent. This is an adaptive defense mechanism for a child with an unsafe caregiver, but maladaptive when the child grows up and seeks to enter into healthy adult relationships. The person with disorganized attachment may experience safe intimacy as stressful, and consequently abandon safe relationships and instead choose partners who reproduce their early environment, e.g., partners who are either avoidant or abusive or both.

The final chapter considers how to prevent or reverse the chronic stress response and adverse behavioral patterns triggered by abusive or neglectful parents. Sarto-Jackson focuses on holistic interventions for at-risk children and adolescents (e.g., the “Circle of Courage” approach described by Brendtro et al., 2005). Sarto-Jackson also highlights the importance of “alloparenting”, in which children receive parenting from those other than their biological parents. In closing, Sarto-Jackson sees her work as a contribution to “psychoeducation”, providing “knowledge at the interface of neuroscience, biological psychology, and social pedagogy” (p. 240) that experts can give to those afflicted by adverse experiences.

A central virtue of Sarto-Jackson’s monograph is its comprehensive illustration of the complexity of developmental processes and the dangers of simplistic explanations. Sarto-Jackson does not just disavow a simplistic “nature versus nurture” dichotomy. She also avoids privileging any particular sets of distinctions or any particular disciplines. In this way, she seems to adhere to what philosophers of science have called the “parity thesis” (P. E. Griffiths & Knight, 1998). The parity thesis not only opposes dividing causal factors influencing development into one set representing natural qualities (e.g., DNA, inborn temperament), and another representing everything else (e.g., the environment). It also highlights the fact that there are many sets of helpful distinctions that can be used to describe the causal factors relevant to developmental explanation. The choice of distinctions used should be contextual; based on the research questions we seek to answer, rather than some notion of metaphysical primacy (Oyama et al., 2003; P. E. Griffiths & Gray, 2005).

Having said this, the parity thesis should not be read to imply that all causal factors are somehow “equally” relevant, or that the most comprehensive explanation is the best explanation. A map with “the scale of a mile to the mile” (Carroll, 1894, p. 169), after all, will have all the relevant details intrepid adventurers require, but will be difficult to carry and perhaps impossible to use (also see Borges, 1964/1975; Donovan & Murphy, 2019). To illustrate, consider these facts, each of which is used by Sarto-Jackson to help explain either Waldroup’s behavior in particular, or abusive behavior more generally. First, Waldroup carries the “warrior gene”, known to predict aggression. Second, those with the “warrior gene” have lowered serotonin levels, which correlates with heightened aggression. Third, Waldroup was abused as a child, and abused children tend to recreate their early social environments as adults. Fourth, traumatized individuals are habituated to live in an “emotionally relabeled” world in which neutral stimuli are perceived as threats. Fifth, this habituation may be adaptive, because it enabled survival in an early hostile environment (and will continue being adaptive if the person continues living in a hostile environment).

Do each of these explanatory components need to be invoked to explain Waldroup’s violent behavior, or is one (or more) doing the “heavy lifting”? There are two issues here. The first is whether each explanation is a necessary component of a larger explanation (a view I take Sarto-Jackson to be endorsing), or whether one or more is “reducible” to another (e.g., do descriptions of psychological function reduce to neurobiological mechanisms?). The question of how different “levels of explanation” relate arises in the broader context of twentieth-century debate about the “unity” of the sciences (see Bechtel & Hamilton, 2007 for a history and survey). One popular strand of twentieth-century thinking has tried to reduce the theories of higher-level sciences to those of lower-level sciences (e.g., Nagel, 1961; Oppenheim & Putnam, 1958). Another strand has rejected the possibility of such reduction (Fodor, 1997; Hull, 1972). More recently, there have been calls to abandon identifying different scientific disciplines with “levels of description” and to instead talk of descriptive levels in the context of localized, multilevel mechanisms (see Piccinini & Craver, 2011 for discussion).

The second issue concerns whether one or more of Sarto-Jackson’s explanatory components are more important than other ones, and whether we should take seriously the idea that “more” is better. Some researchers have argued that medicine is improved when mechanistic explanation (e.g., neurobiological) is supplemented with evolutionary explanation (Nesse & Williams, 1996). There is also the well-known idea that the convergence of two independent lines of inquiry on a similar result provides stronger evidence in its favor (Vicedo, 2009). However, the more comprehensive an explanation, the more unwieldy. This is why diagnostic and etiological frameworks tend to favor parsimony over comprehensiveness.

To balance these explanatory desiderata, Waters’s (2007) concept of causes that “make a difference” might be useful. Consider Mary lighting a match. Mary striking the match causes the match to light. The presence of oxygen in the room also causes the match to light (since, without oxygen, the match would not have lit). Nonetheless, Mary striking the match seems to be the relevant causal factor. Waters calls this factor the “actual difference-maker”. The oxygen is also a cause, but the cause that “makes a difference” is Mary’s lighting the match. (I have given only a metaphorical presentation of this concept. For a definition and discussion see P. E. Griffiths et al., 2015; Waters, 2007).

The notion of a cause (or the cause) doing the heavy lifting in a given context could help make the research questions Sarto-Jackson raises more specific and more tractable. Consider the question: “What predicts a violent temperament in

adulthood?” Sarto-Jackson reports that 33% of the population have the “warrior gene”, but most do not commit violent crimes. Therefore, this gene is not the actual difference-maker in the population at large. A more specific research question could be posed that focused only on the “warrior gene” population and controlled for a history of child abuse. If every person in this narrower population committed violent crimes if they experienced significant childhood abuse, the latter would be the cause that made the difference. However, this finding would be relevant only to the population of “warrior gene” carriers. Actual causal difference-makers may be easier to identify when more variables are held fixed. And identifying them matters not just for parsimonious explanation, but for identifying effective interventions.

In closing, I note that an admirable feature of Sarto-Jackson’s work is its thorough and careful presentation of an intellectual history involving preformationist and determinist ideas about human nature that increasingly give way to greater recognition of the role played by social context in the mid- to late-twentieth century. One notable omission from the book that might be of interest to the author, however, concerns the interesting cross-talk between ethologists and psychoanalysts in the early- to mid-twentieth century. During this period, Bowlby’s (1951) drew on ethologist Lorenz’s work on imprinting in birds to lend support to his work on attachment theory. Imprinting is the process by which some birds become attached to and follow the first moving object they encounter after they hatch (Lorenz was frequently photographed being followed by a brood of goslings). While Bowlby drew on ethology, Lorenz himself drew on emerging work on the mother-child dyad in the work of psychoanalytic thinkers, describing a “progressive synthesis of ethology and psychoanalysis” (Lorenz & Martin, 1969, p. xii).

The mutual support between psychology and ethology was central in popularizing the post-WWII idea that mother-child interactions early in life predicted the quality of emotional temperament in adulthood (Vicedo, 2009). Psychology and psychoanalysis attempted to describe the “internal structures” that led individuals to seek the same social niche over time (like an imprinted bird), while ethology described the instinctive nature of the original attachment an animal forms toward its initial caregiver. In this case, the convergence of two disciplines brought a new idea to prominence. Sarto-Jackson’s monograph, in providing a similar sort of convergences across a larger range of disciplines, may hold the promise of achieving the same. Its explanatory breadth offers an excellent starting-point for formulating more specific and targeted research questions, ones that might be answered by identifying causal “figure and ground” (e.g., difference-making causes versus causes simpliciter) for a given context and set of research questions.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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<https://doi.org/10.1080/09515089.2023.2193242>

